

An Attention-Based Signed Distance Field Estimation Method for Hand-Object Reconstruction

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ABSTRACT

Joint reconstruction of hands and objects from monocular RGB images is a challenging task. In this work, we present a novel hybrid model for joint reconstruction of hands and objects. The model proposed consists of three key modules. Among them, multi-scale attention feature extractor is designed to enhance cross-scale information extraction. Attention-based graph encoder can encode the graph information of the hands. Interacting attention module is applied to fuse information between hands and object. Test results on ObMan dataset [6] show that our method outperforms the state-of-the-art method around 13.97% and 15.75% in H_{se} and H_{je} respectively.

Index Terms: Hand-object reconstruction; Signed distance field; Multi-scale attention feature extractor; Attention-based graph encoder; Interacting attention

1 INTRODUCTION

Reconstruction and tracking of hands and objects using virtual data provides new solutions for human-computer interaction and has the potential for extensive applications and development in the fields of virtual reality, augmented reality, and robotic manipulation. With the development of machine learning, methods for hand and object reconstruction and tracking have made tremendous progress. Yet reconstruction and tracking for detailed interaction processes between hands and objects in hands is still an area which remains further explored.

Some of the previous work focused on reconstruction using 3D data including multi-view images [2], point clouds [1], depth maps [3], etc. To increase the generality and practicality of the proposed method, our research endeavors to use monocular RGB images for hand and object reconstruction. Specifically, we aim to reconstruct and track the detailed interaction process between the hand and the manipulated object from the mutual occlusion of the hand and the object.

Existing methods can be broadly classified into parametric mesh model-based methods and implicit neural representations-based methods. The parametric mesh model-based methods [5,6,9] usually built on MANO [8], which is a commonly used parametric hand model. As MANO contains a prior information from 3D scans of real hands, the model is capable of providing hand reconstruction results that are closer to the real situation. Hasson et al. [6] used the contact loss of the hand and the object to constrain the interaction space of the reconstruction based on the MANO model. However,

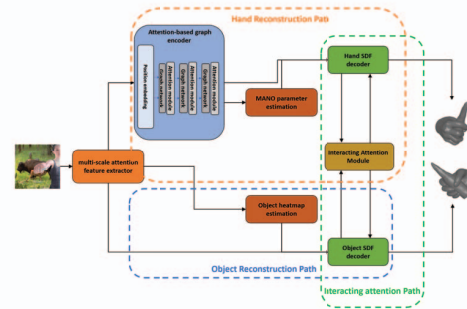


Figure 1: Overview of the our method.

due to the limited shape deformation and mesh resolution of the parametric model, there is still potential to improve the accuracy. Karunratanakul et al. [7] proposed to represent the hand and object models and interaction results in a unified signed distance field(SDF), which has shown great potential in recent years in reconstruction and scene representation. A hybrid approach [4] was proposed to estimate the pose of the hand and object based on the parametric model firstly, and then learn the SDF of the hand and object based on the estimated pose to complete the reconstruction of the hand-object interaction.

Although the hybrid approach encompasses the advantage of the first two approach, there is still potential to improve in the current jointed methods. The previous convolution-based feature extractor can hardly satisfy the requirements of mining global information for MANO's modeling and local information for SDF estimation; for this reason, we introduce a multi-scale network to enhance cross-scale information extraction. As the representation of the hand is a natural graph, an attention-based graph encoder is used to encode the features of the hands. In addition, the parametric learning of pose and shape by the individual decoders may not be sufficient for the construction during hand-object interaction where the hand and object are learned disentangled. To address this, an interactive attention module is applied to perform information fusion interaction between hands and object to improve the prediction accuracy of the interaction process.

2 METHODS

Building on existing work, a multi-scale attention feature extractor is used to improve the feature extraction ability of the network. Besides, attention-based graph module and interacting attention module are added to the network to improve the performance of our method.

The multi-scale attention feature extractor can well solve the problem of association between large-scale global information and small-scale fine-grained information. It is ideal for tasks that require

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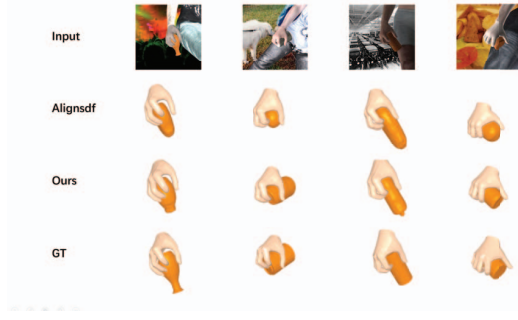


Figure 2: Hand and object reconstruction.

pose estimation and surface reconstruction at the same time.

$$F_M^1, F_M^2, F_M^3 = f_M(input) \quad (1)$$

where $input$ denotes monocular rgb image, F_M^1, F_M^2, F_M^3 represent feature maps for different paths respectively.

For hand pose estimation and reconstruction task, we still use MANO as a basic framework. The attention-based graph encoder is divided into two paths, one path is used to encode the shape features of hand, and the other path is used to regress the parameters of MANO. The attention-based graph encoder includes a graph convolution module and an attention module. The graph convolution module is designed to fuse the feature map from the feature extractor and the prior information of the joint connection of the hand. The attention module is used to fuse the information of the two pathways, so that the hand pose estimation and reconstruction tasks can promote each other. The attention-based graph encoder can better regress the MANO parameters and extract features related to hand shape, and then estimate the pose according to the MONA parameters. The pose estimation result will be send to the hand SDF decoder together with the previous feature map to calculate the signed distance.

$$F_A^1, F_A^2 = f_A(F_M^1) \quad (2)$$

where f_A denotes the attention-based graph encoder, F_M^2 represents feature map for MANO parameter estimating, F_M^1 is the feature map input to the hand SDF decoder.

The interacting attention module between the SDF decoder of the hand and the SDF decoder of the object can provide a way of interacting between hand and object, enabling both of them to estimate the SDF with the help of each other's information.

$$D_{inter}^H, D_{inter}^O = f_{inter} \left(f_{mano} \left(F_A^2 \right) \oplus F_A^1, f_{heat} \left(F_M^2 \right) \oplus F_M^3 \right) \quad (3)$$

where D_{inter}^H and D_{inter}^O denote signed distance estimation results of hand and object, f_{inter}^H denotes the interacting attention module, f_{mano}^H represents MANO parameter estimation module, f_{heat}^H is object position estimation module.

3 EXPERIMENTS

Preliminary experiment was conducted to verify the effectiveness of the multi-scale attention feature extractor and interacting attention module. Due to the extremely large size of the dataset, the preliminary experiment was trained on 70k data selected data in the dataset randomly.

Model	train data	H _{se}	H _{je}	O _{te}
Hasson <i>et al.</i> [6]	87k	0.415	1.13	-
Karunratanakul <i>et al.</i> [7]-1De	87k	0.261	-	-
Karunratanakul <i>et al.</i> [7]-2De	87k	0.237	-	-
AlignSDF [4]	87k	0.136	1.27	3.29
Ours	70k	0.117	1.07	3.32

Table 1: Comparison results with the state-of-the-art method on the ObMan dataset.

In the experiment, the feature extraction module in [4] was replaced with the aforementioned multi-scale attention feature extractor and the interacting attention module was added. As shown in table 1, the results of the comparison show that our method reduces the error of the hand reconstruction by 0.012 and 0.20 in H_{se} and H_{je} respectively. H_{se}(cm²) is the median chamfer distance over the entire test set. H_{je}(cm) is the mean joint error relative to the hand wrist joint over 21 joints.

Further experiments were conducted to evaluate the performance of interacting attention module. The comparison results show that the loss of the model with the interacting attention module decreased more significantly and the model converged faster during the train procedure.

4 CONCLUSION

We proposed a novel hand object reconstruction method. This method consists of multi-scale attention feature extractor, attention-based graph encoder, and interacting attention module. In order to verify our method, we conducted some experiments and obtained a relatively competitive result compared with the state-of-the-art methods.

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